

- Students will perform security posture assessment
 - Students will use OSINT to gather information
 - Students will use Google dorks for performing various operations
 - Students will use shodan.io for analysis
-

1. Defining Scope and Objectives

Purpose: Establish boundaries and expectations before any technical activity.

- Practical Actions
 - Identify assets in scope:
 - Servers (Windows/Linux)
 - Network devices (firewalls, routers, switches)
 - Applications (web, internal apps), Cloud resources (if applicable)
 - Define assessment type:
 - Configuration review
 - Vulnerability assessment
 - Compliance assessment
 - Full security posture assessment

2. Asset Discovery and Inventory

Purpose: You cannot secure what you do not know.

- Practical Actions
 - Identify live assets using:
 - `nmap -sn <network-range>`
 - Active Directory computer objects
 - Cloud dashboards
 - Classify assets:
 - Critical / Non-critical
 - Production / Test / Dev
- Tools
 - Nmap, Active Directory
 - CMDB / Excel
 - Cloud inventory tools

3. Baseline Security Controls Review

Purpose: Measure current controls against best practices.

- Areas to Review
 - Authentication & Authorization
 - Patch management
 - Backup & recovery
 - Logging & monitoring
 - Network segmentation
- Practical Checks
 - Password policy in AD (secpol.msc → local password policy editor.)
 - Firewall rules review (Palo Alto / Windows Firewall)
 - Antivirus/EDR presence
 - Backup schedules and restore testing

4. Configuration Assessment (Hardening Check)

Purpose: Identify insecure system and network configurations.

Practical Actions

- Windows
 - Check:
 - RDP restrictions
 - Local admin usage
 - Tools:
 - Microsoft Security Baseline
 - `gpresult /r`

- Linux
 - Check:
 - SSH root login
 - File permissions
 - Firewall rules (firewalld, iptables)
 - Commands:
 - `sudo ss -tuln`
 - `sudo firewall-cmd --list-all`
- Network Devices
 - Default credentials
 - Unused open ports
 - Weak encryption protocols

5. Vulnerability Assessment

Purpose: Identify known exploitable weaknesses.

Practical Steps

- Deploy a vulnerability scanner
- Configure scan policy
- Run authenticated scans (preferred)
- Analyze results

Tools

- Nessus, Qualys, OpenVAS

Identify:

- Critical CVEs ((Common Vulnerabilities and Exposures)
- Missing patches
- Weak ciphers

6. Network Security Assessment

Purpose: Validate network-level protection.

Practical Checks

- Firewall rule base review
- NAT and segmentation validation
- IDS/IPS status
- Open ports and exposed services
- Commands
 - `nmap -sV -p- <target-ip>`
 - `tracert <destination>`

Now Practical:

Lab requirement

- 1 linux server (192.168.88.155/24)
 - Internet active: yes
- 1 windows server (192.168.88.158/24)
 - Internet active: yes
- Using NMAP as CLI for tracing details on linux.
- Installing & Using Nessus web-based tool for gathering information about target.

On linux server:

```
[root@server ~]# nmap -sn 192.168.88.0/24
Starting Nmap 7.92 ( https://nmap.org ) at 2026-01-07 22:49 IST
Nmap scan report for 192.168.88.1
Host is up (0.00036s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.88.2
Host is up (0.00014s latency).
MAC Address: 00:50:56:F4:DC:E9 (VMware)
Nmap scan report for 192.168.88.155
Host is up (0.00018s latency).
MAC Address: 00:0C:29:49:A3:C0 (VMware)
Nmap scan report for 192.168.88.254
Host is up (0.00014s latency).
MAC Address: 00:50:56:FE:45:49 (VMware)
Nmap scan report for 192.168.88.151
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.03 seconds
[root@server ~]#
```

Command

- “-sn” – it checks only hosts are up or not. It sends ICMP echo ping request to port 443.

Scanning port and service using “nmap -sS -sV -O 192.168.88.155”

```
[root@server ~]# nmap -sS -sV -O 192.168.88.155
Starting Nmap 7.92 ( https://nmap.org ) at 2026-01-08 11:49 IST
Nmap scan report for 192.168.88.155
Host is up (0.00025s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
80/tcp    open  http           Microsoft IIS httpd 10.0
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds   Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
MAC Address: 00:0C:29:49:A3:C0 (VMware)
Device type: general purpose
Running: Microsoft Windows 2016
OS CPE: cpe:/o:microsoft:windows_server_2016
OS details: Microsoft Windows Server 2016 build 10586 - 14393
Network Distance: 1 hop
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.47 seconds
```

Command:

- “-sS” = tcp syn scan
- “-sV” = version detection
- “-O” = detect OS version

After installing DNS role on Windows server, run same command “nmap -sS -sV -O 192.168.88.156”

```
[root@server ~]# nmap -sS -sV -O 192.168.88.155
Starting Nmap 7.92 ( https://nmap.org ) at 2026-01-08 12:13 IST
Nmap scan report for 192.168.88.155
Host is up (0.00034s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Simple DNS Plus
80/tcp    open  http           Microsoft IIS httpd 10.0
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds   Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
MAC Address: 00:0C:29:49:A3:C0 (VMware)
Device type: general purpose
Running: Microsoft Windows 2016
OS CPE: cpe:/o:microsoft:windows_server_2016
OS details: Microsoft Windows Server 2016 build 10586 - 14393
Network Distance: 1 hop
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows
```

Installing Nessus for networking monitoring:

Download link: <https://www.tenable.com/downloads/nessus?loginAttempted=true>

Selecting correct version:

1 Download and Install Nessus

Choose Download

Version

Nessus - 10.11.1



Platform

Linux - RHEL 8 - x86_64

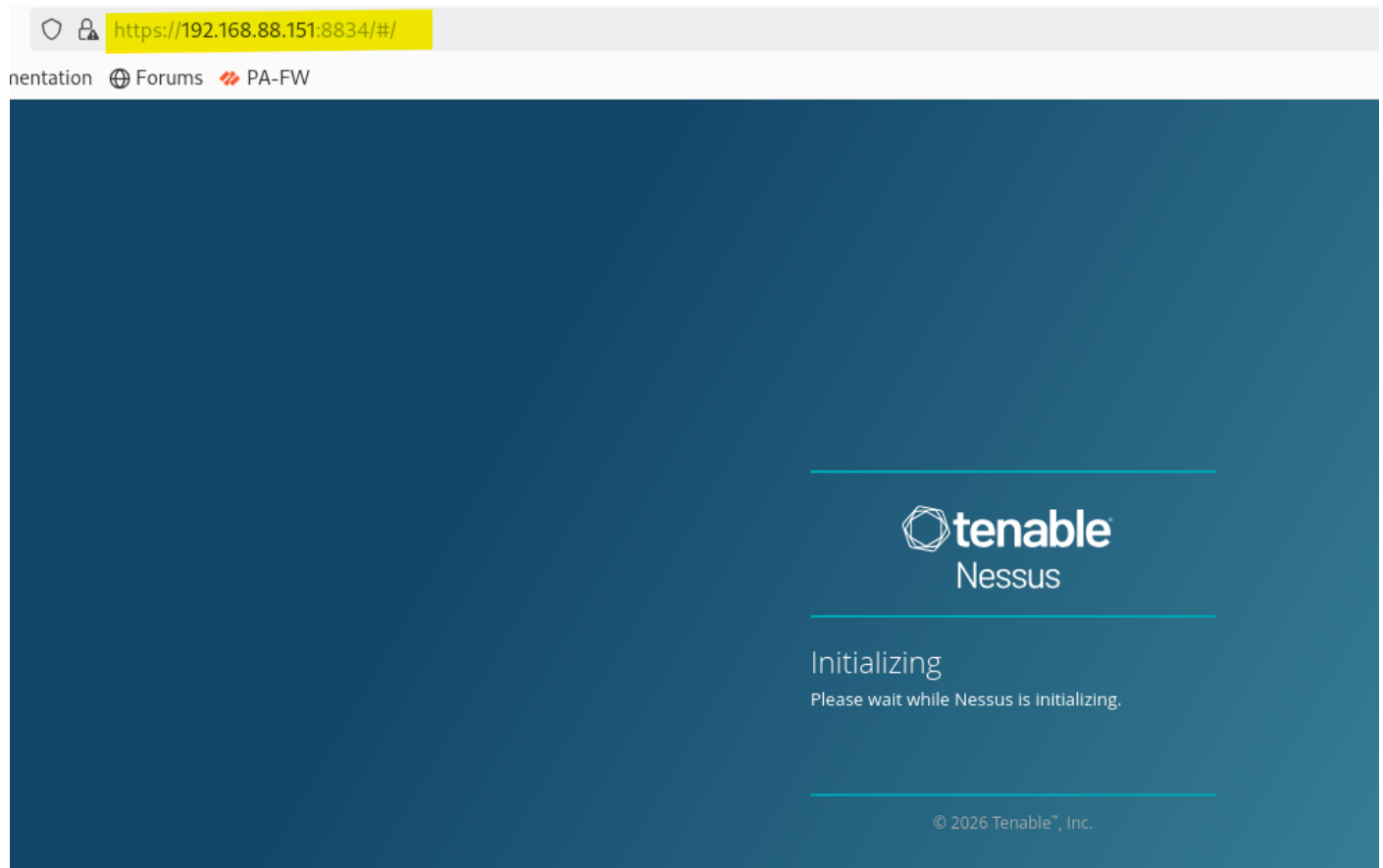


Download

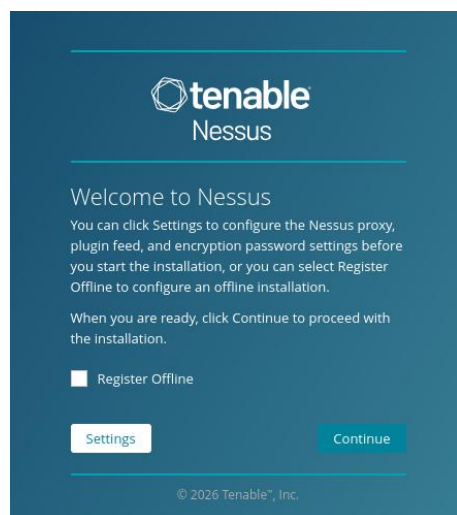
Checksum

- `sudo systemctl start nessusd`
- `sudo systemctl enable nessusd`

accessing Nessus on firefox <https://192.168.88.151:8834/>



Register Nessus Essential for small network



tenable
Nessus

Welcome to Nessus

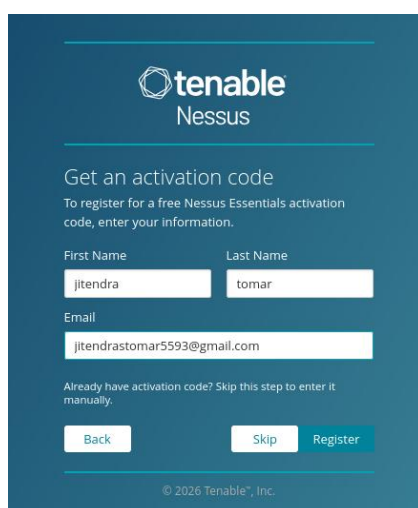
You can click Settings to configure the Nessus proxy, plugin feed, and encryption password settings before you start the installation, or you can select Register Offline to configure an offline installation.

When you are ready, click Continue to proceed with the installation.

☐ Register Offline

[Settings](#) [Continue](#)

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Nessus

Get an activation code

To register for a free Nessus Essentials activation code, enter your information.

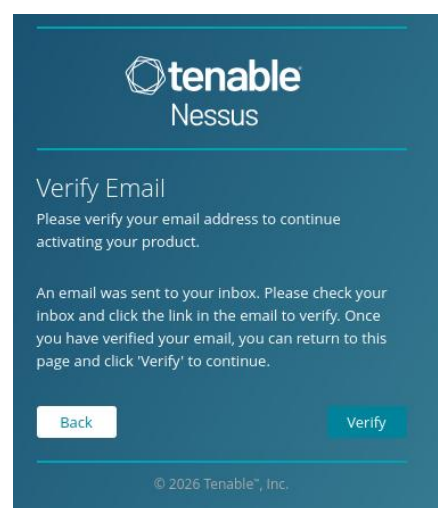
First Name Last Name

Email

Already have activation code? Skip this step to enter it manually.

[Back](#) [Skip](#) [Register](#)

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Nessus

Verify Email

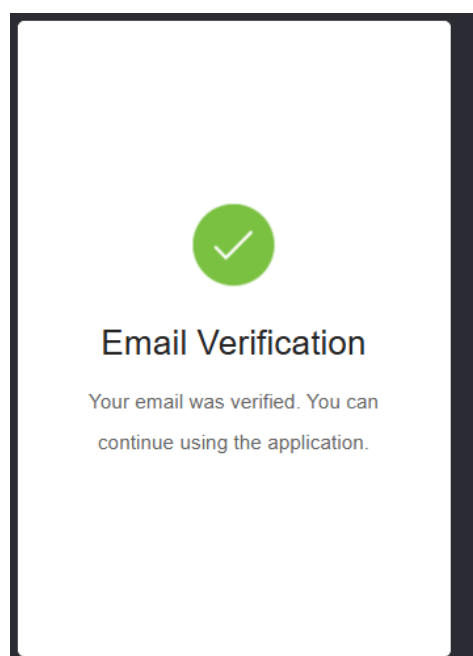
Please verify your email address to continue activating your product.

An email was sent to your inbox. Please check your inbox and click the link in the email to verify. Once you have verified your email, you can return to this page and click 'Verify' to continue.

[Back](#) [Verify](#)

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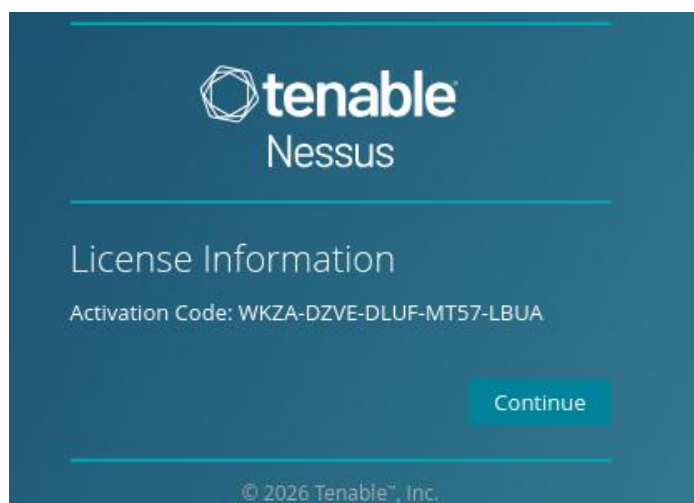
Verify over your email ID and click on verify button.





Email Verification

Your email was verified. You can continue using the application.



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Nessus

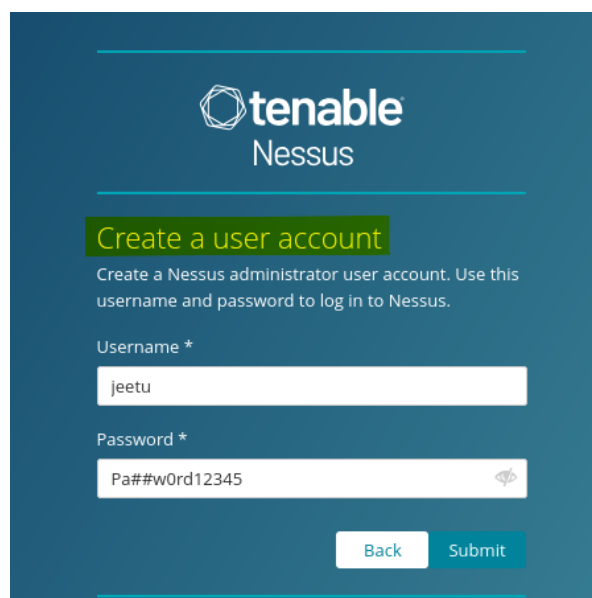
License Information

Activation Code: WKZA-DZVE-DLUF-MT57-LBUA

[Continue](#)

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Activation Code: WKZA-DZVE-DLUF-MT57-LBUA



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Nessus

Create a user account

Create a Nessus administrator user account. Use this username and password to log in to Nessus.

Username *

Password *

[Back](#) [Submit](#)



tenable
Nessus

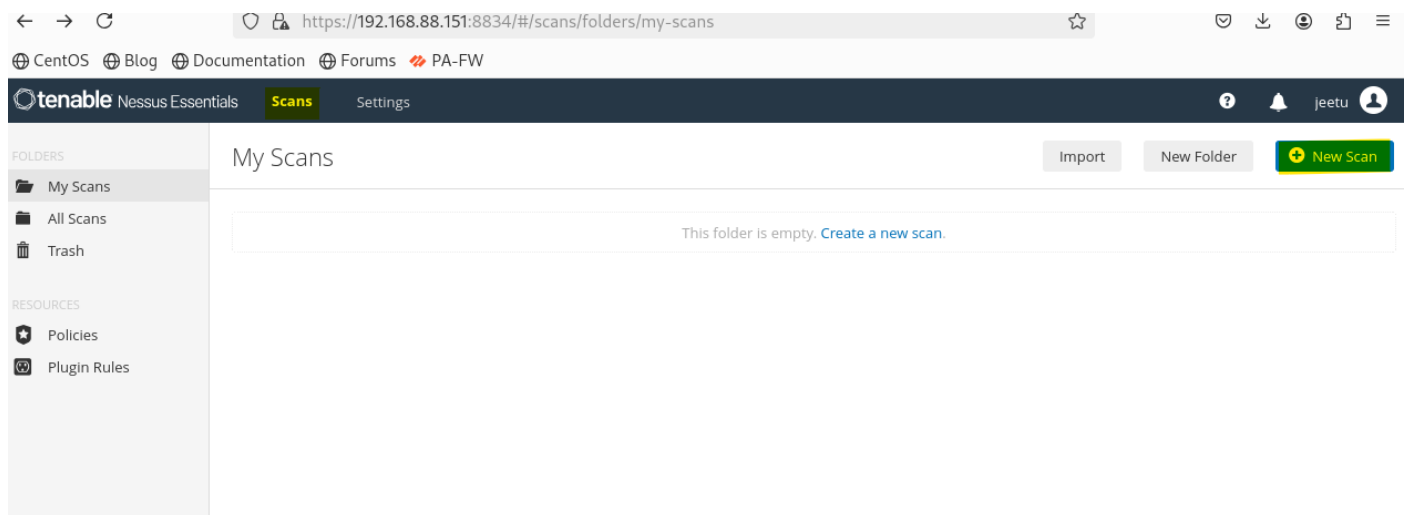
Initializing

Please wait while Nessus is initializing.

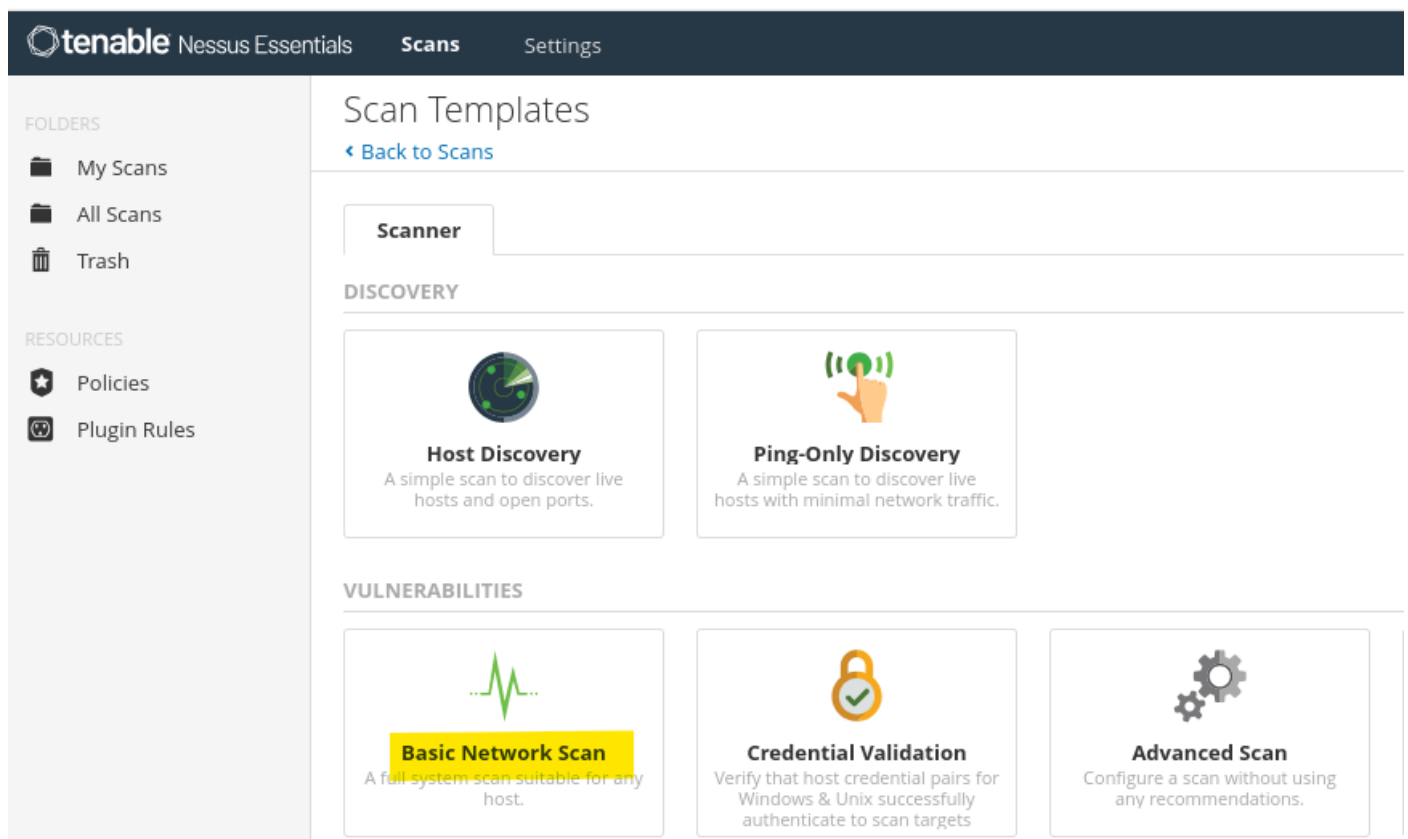
Downloading plugins...

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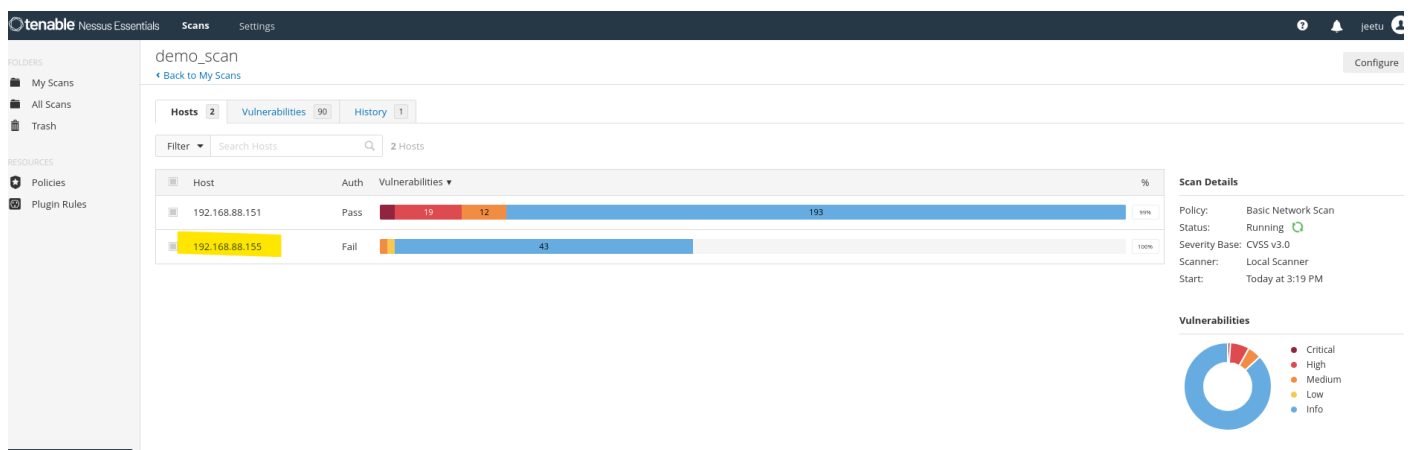
Goto Scan → New Scan



Choose Basic Network Scan



After scanning,



Once you click the target IP address, it will list all the vulnerabilities:

The screenshot shows the Tenable Nessus Essentials interface. The top navigation bar includes 'tenable', 'Nessus Essentials', 'Scans', and 'Settings'. The left sidebar has 'FOLDERS' (My Scans, All Scans, Trash) and 'RESOURCES' (Policies, Plugin Rules). The main content area is titled 'demo_scan / 192.168.88.155' with a 'Configure' button. Below the title is a 'Vulnerabilities' section showing 22 vulnerabilities. A table lists these vulnerabilities with columns for Severity, CVSS, VPR, EPSS, Name, Family, and Count. The table includes entries like 'SMB (Multiple Issues)', 'ICMP Timestamp Request Remote Date Disclosure', 'HTTP (Multiple Issues)', 'DCE Services Enumeration', 'Nessus SYN scanner', 'Service Detection', 'Common Platform Enumeration (CPE)', 'Device Type', 'Ethernet Card Manufacturer Detection', and 'Ethernet MAC Addresses'. A 'Host Details' sidebar on the right shows IP, MAC, OS, Start, End, Elapsed, and Auth information. A 'Vulnerabilities' donut chart is also present.

Sev	CVSS	VPR	EPSS	Name	Family	Count
MIXED	...	...	...	SMB (Multiple Issues)	Misc.	2
LOW	2.1 *	...	...	ICMP Timestamp Request Remote Date Disclosure	General	1
INFO	...	...	...	SMB (Multiple Issues)	Windows	6
INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	5
INFO	...	...	...	DCE Services Enumeration	Windows	9
INFO	...	...	...	Nessus SYN scanner	Port scanners	5
INFO	...	...	...	Service Detection	Service detection	2
INFO	...	...	...	Common Platform Enumeration (CPE)	General	1
INFO	...	...	...	Device Type	General	1
INFO	...	...	...	Ethernet Card Manufacturer Detection	Misc.	1
INFO	...	...	...	Ethernet MAC Addresses	General	1

Click on vulnerabilities

The screenshot shows the Tenable Nessus Essentials interface. The top navigation bar includes 'tenable', 'Nessus Essentials', 'Scans', and 'Settings'. The left sidebar has 'FOLDERS' (My Scans, All Scans, Trash) and 'RESOURCES' (Policies, Plugin Rules). The main content area is titled 'demo_scan' with a 'Back to My Scans' link. Below the title is a 'Vulnerabilities' section showing 90 vulnerabilities. A table lists these vulnerabilities with columns for Severity, CVSS, VPR, EPSS, Name, Family, and Count. The table includes entries like 'CentOS (Multiple Issues)', 'Nagios Xi (Multiple Issues)', 'NFS Shares World Readable', 'Apache Tomcat (Multiple Issues)', 'SSL (Multiple Issues)', 'SMB (Multiple Issues)', 'ICMP Timestamp Request Remote Date Disclosure', 'HTTP (Multiple Issues)', 'SSH (Multiple Issues)', 'SMB (Multiple Issues)', and 'Mysql (Multiple Issues)'. A 'Scan Details' sidebar on the right shows Policy, Status, Severity Base, Scanner, and Start information. A 'Vulnerabilities' donut chart is also present.

Sev	CVSS	VPR	EPSS	Name	Family	Count
MIXED	...	...	...	CentOS (Multiple Issues)	CentOS Local Security Checks	22
MIXED	...	...	...	Nagios Xi (Multiple Issues)	CGI abuses	4
HIGH	7.5	...	...	NFS Shares World Readable	RPC	1
MIXED	...	...	...	Apache Tomcat (Multiple Issues)	Web Servers	8
MIXED	...	...	...	SSL (Multiple Issues)	General	10
MIXED	...	...	...	SMB (Multiple Issues)	Misc.	2
LOW	2.1 *	...	...	ICMP Timestamp Request Remote Date Disclosure	General	1
INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	16
INFO	...	...	...	SSH (Multiple Issues)	General	8
INFO	...	...	...	SMB (Multiple Issues)	Windows	6
INFO	...	...	...	Mysql (Multiple Issues)	Databases	4

What Is Credentialed Scanning (Windows)?

Credentialed scanning allows Nessus to **log in to the Windows system** and perform **deep inspection**, including:

- Missing security patches
- Local vulnerabilities
- Weak configurations
- Installed software inventory
- Registry and policy checks

This provides **far more accurate results** than unauthenticated scans.

Settings → Credentials → Windows

Credential Options

Fill in:

- **Username:** Administrator or DOMAIN\user
- **Password:** Admin password
- **Authentication method:** Password
- **Elevate privileges:** Enabled

Optional:

- Enable **SMB Domain**
- Enable **Start Remote Registry service**

Step 3: Configure the Scan

1. Go to **Scans → New Scan**
2. Select **Basic Network Scan**
3. Enter target IP or hostname
4. Go to **Credentials tab**
5. Add **Windows credentials**
6. Save the scan

Step 8: What Credentialed Scanning Finds That Non-Credentialed Cannot

Area	Non-Cred	Credentialed
Missing patches	✗	✓
Local policies	✗	✓
Registry issues	✗	✓
Installed software	✗	✓
Accurate severity	✗	✓

Students will use OSINT to gather information

OSINT – Open-Source Intelligence, refers to the practice of collecting and analysing publicly available information to identify threats, assess risks and strengthen defences.

Tools

- Using WHOIS database on wipro.com domain.

```
(kali@kali)-[~]
$ whois wipro.com
Domain Name: WIPRO.COM
Registry Domain ID: 2559888_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.registrar.amazon
Registrar URL: http://registrar.amazon.com
Updated Date: 2025-10-28T11:30:36Z
Creation Date: 1992-12-08T05:00:00Z
Registry Expiry Date: 2030-12-07T05:00:00Z
Registrar: Amazon Registrar, Inc.
Registrar IANA ID: 468
Registrar Abuse Contact Email: trustandsafety@support.aws.com
Registrar Abuse Contact Phone: +1.2024422253
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS-1145.AWSDNS-15.ORG
Name Server: NS-1575.AWSDNS-04.CO.UK
Name Server: NS-326.AWSDNS-40.COM
Name Server: NS-659.AWSDNS-18.NET
DNSSEC: signedDelegation
DNSSEC DS Data: 31918 13 2 0074B6D159450F1E12E2243EFEA93235A69CBDB77A1CD6A795EEA4F87E7A5C9E
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-01-08T14:28:14Z <<<
```

Other useful information:

```
Registrant City: Hayes
Registrant State/Province: Middlesex
Registrant Postal Code: UB3 9TR
Registrant Country: GB
Registrant Phone: +44.1483307527
Registrant Phone Ext:
Registrant Fax: +44.1483304031
Registrant Fax Ext:
Registrant Email: 7649f398-8b17-4de3-9427-9d9020cca728@identity-protect.org
Registry Tech ID: Not Available From Registry
Tech Name: On behalf of wipro.com owner
Tech Organization: Identity Protection Service
Tech Street: PO Box 786
Tech City: Hayes
Tech State/Province: Middlesex
Tech Postal Code: UB3 9TR
Tech Country: GB
Tech Phone: +44.1483307527
Tech Phone Ext:
Tech Fax: +44.1483304031
Tech Fax Ext:
Tech Email: 7649f398-8b17-4de3-9427-9d9020cca728@identity-protect.org
Name Server: NS-659.AWSDNS-18.NET
Name Server: NS-326.AWSDNS-40.COM
Name Server: NS-1575.AWSDNS-04.CO.UK
Name Server: NS-1145.AWSDNS-15.ORG
```

Using dig command for domain Wipro.com for fetching Public IP address.

```
(kali㉿kali)-[~]
$ dig wipro.com

; <>> DiG 9.20.9-1-Debian <>> wipro.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 63955
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 4096
;; QUESTION SECTION:
;wipro.com.                IN      A

;; ANSWER SECTION:
wipro.com.                5       IN      A      18.205.78.10

;; Query time: 44 msec
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)
;; WHEN: Thu Jan 08 09:30:46 EST 2026
;; MSG SIZE rcvd: 54
```

Verifying it using PING:

```
C:\Users\Jeetu>ping wipro.com

Pinging wipro.com [18.205.78.10] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 18.205.78.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Using DNSDUMPSTER to gather global information:

A Records (subdomains from dataset)					
Host	IP	ASN	ASN Name	Open Services (from DB)	RevIP
apiaccountservices.wipro.com	203.91.197.71	ASN: 23664 203.91.197.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1
autodiscover.wipro.com	203.91.193.61	ASN: 23664 203.91.193.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1
azeu-p-mcafagnt.wipro.com	52.224.135.151	ASN: 8075 52.224.0.0/11	MICROSOFT-CORP-MSN-AS-BLOCK United States		1
base.wipro.com	203.91.193.54	ASN: 23664 203.91.193.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1
blrweb-ext.wipro.com	203.91.193.123	ASN: 23664 203.91.193.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1
cellentvpn.wipro.com	62.154.234.100	ASN: 3320 62.154.0.0/15	DTAG Internet service provider operations, DE Germany		1
cloudcar.wipro.com	3.136.185.207 ec2-3-136-185-207.us- east-2.compute.amazonaws.	ASN: 16509 3.136.0.0/13	AMAZON-02 United States		2

1. Students will use Google dorks for performing various operations

- Google Dorks (or Google hacking queries) are advanced search operators used to uncover hidden information on websites.
- They leverage Google's indexing to find files, misconfigurations, or sensitive data that shouldn't be publicly accessible.

Google Dork: **filetype:pdf site:wipro.com**

filetype:pdf site:wipro.com

×

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🔍

AI Mode **All** Shopping Images Short videos Videos Forums More ▾ Tools ▾

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HEALTH AND SAFETY POLICY

15 Jul 2024 — At Wipro, the well-being and safety of our employees is of utmost importance. Well-being refers to a state characterized by physical and ... [Read more](#)

7 pages

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WIPRO INSURANCE SOLUTIONS, LLC

28 May 2024 — Opinion. We have audited the accompanying financial statements of Wipro Insurance Solutions, LLC, which are comprised of the balance sheets ... [Read more](#)

19 pages

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HealthPlan Services Insurance Agency, LLC

31 May 2022 — Opinion. We have audited the accompanying financial statements of HealthPlan Services Insurance Agency, LLC,. [Read more](#)

 Wipro
https://www.wipro.com › nexus › subsidiaries PDF ⓘ

HealthPlan Services Insurance Agency, LLC

27 Mar 2023 — Opinion. We have audited the accompanying financial statements of HealthPlan

Another example of Google Dork

intitle:"index of" "admin"

intitle:"index of" "admin"

×

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AI Mode **All** Images Shopping Videos Short videos News More ▾ Tools ▾

 csmss.org
https://csmss.org › admin › uploads › student ⓘ

Index of /admin/uploads/student

Index of /admin/uploads/student. Name · Last modified · Size · Description · Parent Directory, -. file/, 2026-01-01 09:03, -. image/, 2023-11-29 14:46, - [Read more](#)

Index of /admin/uploads/student/file

Name	Last modified	Size	Description
Parent Directory	-	-	-
2-student.pdf	2023-11-25 18:15	26K	
3-student.pdf	2023-11-25 18:15	26K	
4-student.pdf	2023-11-25 18:14	756K	
10-student.jpg	2023-11-25 18:17	289K	
10-student.pdf	2023-11-25 18:17	397K	
11-student.pdf	2023-11-25 18:17	307K	

Using Exiftool to get the information of an image, video or PDFs.

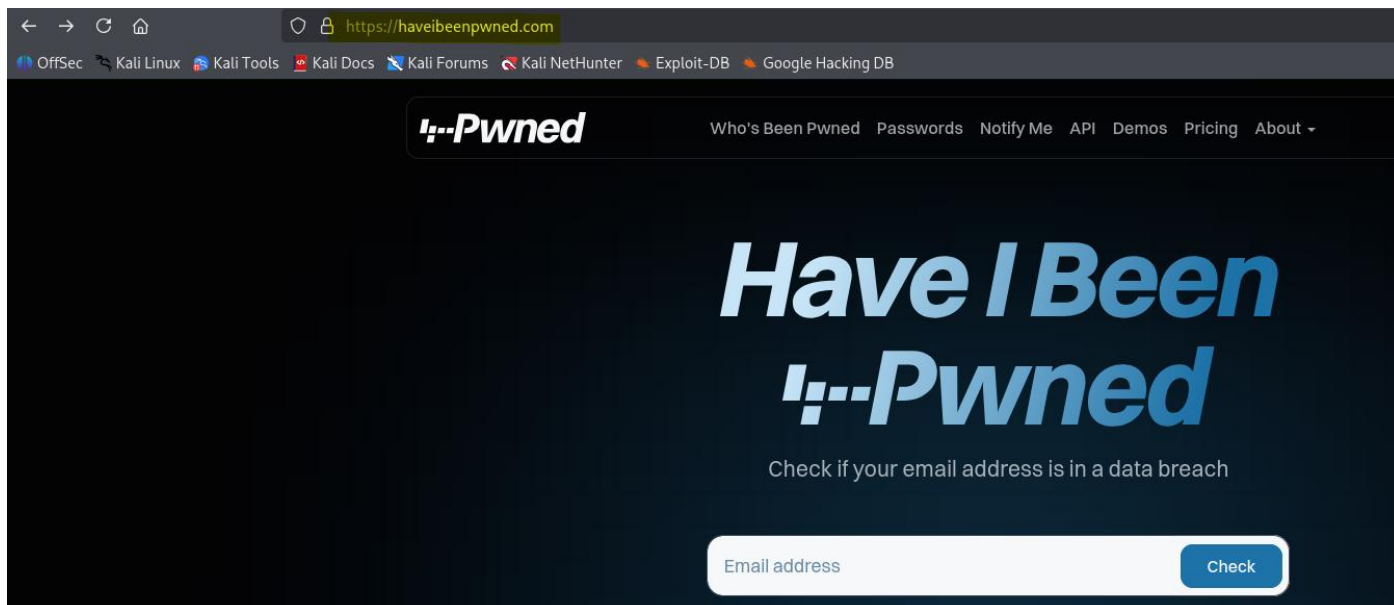
- First you need to download any image from internet. (ex - <https://melangeoftales.com/wp-content/uploads/2024/11/AdiyogiAarthy2-2.jpeg>)
- Then go to the downloaded location and use "exiftool <filename>.extension"

```
(kali㉿kali)-[~/Downloads]
└─$ exiftool AdiyogiAarthy2-2.jpeg
ExifTool Version Number      : 13.25
File Name                    : AdiyogiAarthy2-2.jpeg
Directory                    : .
File Size                     : 144 kB
File Modification Date/Time   : 2026:01:08 10:02:36-05:00
File Access Date/Time        : 2026:01:08 10:02:36-05:00
File Inode Change Date/Time   : 2026:01:08 10:02:36-05:00
File Permissions              : -rw-rw-r--
File Type                     : JPEG
File Type Extension          : jpg
MIME Type                     : image/jpeg
JFIF Version                  : 1.01
Resolution Unit               : None
X Resolution                   : 72
Y Resolution                   : 72
Image Width                   : 1024
Image Height                   : 768
Encoding Process               : Progressive DCT, Huffman coding
Bits Per Sample                : 8
Color Components               : 3
Y Cb Cr Sub Sampling          : YCbCr4:2:0 (2 2)
Image Size                    : 1024x768
Megapixels                    : 0.786
```

I have tried it on an image that I created using Adobe Photoshop CS6

```
(kali㉿kali)-[~/Desktop]
└─$ exiftool jeetu-creative3.jpg
ExifTool Version Number      : 13.25
File Name                    : jeetu-creative3.jpg
Directory                    : .
File Size                     : 1275 kB
File Modification Date/Time   : 2022:08:20 01:25:14-04:00
File Access Date/Time        : 2026:01:08 10:07:26-05:00
File Inode Change Date/Time   : 2026:01:08 10:07:25-05:00
File Permissions              : -rw-----
File Type                     : JPEG
File Type Extension          : jpg
MIME Type                     : image/jpeg
Exif Byte Order               : Big-endian (Motorola, MM)
Photometric Interpretation    : RGB
Orientation                   : Horizontal (normal)
```

You can check if an email ID has been ever breached or not using “<https://haveibeenpwned.com/>”



The image shows a web browser window with the URL <https://haveibeenpwned.com/> in the address bar. The browser's bookmark bar contains links to OffSec, Kali Linux, Kali Tools, Kali Docs, Kali Forums, Kali NetHunter, Exploit-DB, and Google Hacking DB. The website's header features the 'Have I Been Pwned' logo and a navigation menu with links to 'Who's Been Pwned', 'Passwords', 'Notify Me', 'API', 'Demos', 'Pricing', and 'About'. The main content area has a dark background with the text 'Have I Been Pwned' in large, bold, light blue letters. Below this, it says 'Check if your email address is in a data breach'. At the bottom, there is a white input field labeled 'Email address' and a blue 'Check' button.

Have I Been Pwned

Who's Been Pwned Passwords Notify Me API Demos Pricing About ▾

Have I Been Pwned

Check if your email address is in a data breach

Email address

Check

Students will use shodan.io for analysis

- Shodan.io is often called the “search engine for devices” because it indexes information about internet-connected systems rather than websites.
- In cybersecurity, especially in **ethical hacking and penetration testing**, it’s a powerful OSINT tool.
 - o Web search engines, such as Google and Bing, are great for finding websites.
 - o But what if you're interested in measuring which countries are becoming more connected?
 - o Or if you want to know which version of Microsoft IIS is the most popular?
 - o Or you want to find the control servers for malware?
 - o Maybe a new vulnerability came out and you want to see how many hosts it could affect?

Set up your environment

- Create a **Shodan account** (free tier is enough for basic queries).
- Use a **test domain or IP range you own** (or a lab VM with internet exposure).

Query: ‘port:80 country:IN org:"amazon"’

The screenshot shows the Shodan search interface. The search bar contains the query 'port:80 country:IN org:"amazon"'. The results show 320,542 total results. The top cities are listed: Mumbai (309,014), Hyderabad (8,383), Kolkata (3,140), and Delhi (4). The product spotlight mentions a new API for Fast Vulnerability Lookups. The main result is for IP 3.7.99.76, which is an Amazon Data Services India server in Mumbai, India. The result details include the IP, location, and a list of open ports (80, 443, 8080, 8443, 9000, 9443, 9500, 9501, 9502, 9503, 9504, 9505, 9506, 9507, 9508, 9509, 9510, 9511, 9512, 9513, 9514, 9515, 9516, 9517, 9518, 9519, 9520, 9521, 9522, 9523, 9524, 9525, 9526, 9527, 9528, 9529, 9530, 9531, 9532, 9533, 9534, 9535, 9536, 9537, 9538, 9539, 9540, 9541, 9542, 9543, 9544, 9545, 9546, 9547, 9548, 9549, 9550, 9551, 9552, 9553, 9554, 9555, 9556, 9557, 9558, 9559, 9560, 9561, 9562, 9563, 9564, 9565, 9566, 9567, 9568, 9569, 9570, 9571, 9572, 9573, 9574, 9575, 9576, 9577, 9578, 9579, 9580, 9581, 9582, 9583, 9584, 9585, 9586, 9587, 9588, 9589, 9590, 9591, 9592, 9593, 9594, 9595, 9596, 9597, 9598, 9599, 9600, 9601, 9602, 9603, 9604, 9605, 9606, 9607, 9608, 9609, 9610, 9611, 9612, 9613, 9614, 9615, 9616, 9617, 9618, 9619, 9620, 9621, 9622, 9623, 9624, 9625, 9626, 9627, 9628, 9629, 9630, 9631, 9632, 9633, 9634, 9635, 9636, 9637, 9638, 9639, 9640, 9641, 9642, 9643, 9644, 9645, 9646, 9647, 9648, 9649, 9650, 9651, 9652, 9653, 9654, 9655, 9656, 9657, 9658, 9659, 9660, 9661, 9662, 9663, 9664, 9665, 9666, 9667, 9668, 9669, 9670, 9671, 9672, 9673, 9674, 9675, 9676, 9677, 9678, 9679, 9680, 9681, 9682, 9683, 9684, 9685, 9686, 9687, 9688, 9689, 9690, 9691, 9692, 9693, 9694, 9695, 9696, 9697, 9698, 9699, 9700, 9701, 9702, 9703, 9704, 9705, 9706, 9707, 9708, 9709, 9710, 9711, 9712, 9713, 9714, 9715, 9716, 9717, 9718, 9719, 9720, 9721, 9722, 9723, 9724, 9725, 9726, 9727, 9728, 9729, 9730, 9731, 9732, 9733, 9734, 9735, 9736, 9737, 9738, 9739, 9740, 9741, 9742, 9743, 9744, 9745, 9746, 9747, 9748, 9749, 9750, 9751, 9752, 9753, 9754, 9755, 9756, 9757, 9758, 9759, 9760, 9761, 9762, 9763, 9764, 9765, 9766, 9767, 9768, 9769, 9770, 9771, 9772, 9773, 9774, 9775, 9776, 9777, 9778, 9779, 9780, 9781, 9782, 9783, 9784, 9785, 9786, 9787, 9788, 9789, 9790, 9791, 9792, 9793, 9794, 9795, 9796, 9797, 9798, 9799, 9800, 9801, 9802, 9803, 9804, 9805, 9806, 9807, 9808, 9809, 9810, 9811, 9812, 9813, 9814, 9815, 9816, 9817, 9818, 9819, 9820, 9821, 9822, 9823, 9824, 9825, 9826, 9827, 9828, 9829, 9830, 9831, 9832, 9833, 9834, 9835, 9836, 9837, 9838, 9839, 9840, 9841, 9842, 9843, 9844, 9845, 9846, 9847, 9848, 9849, 9850, 9851, 9852, 9853, 9854, 9855, 9856, 9857, 9858, 9859, 9860, 9861, 9862, 9863, 9864, 9865, 9866, 9867, 9868, 9869, 9870, 9871, 9872, 9873, 9874, 9875, 9876, 9877, 9878, 9879, 9880, 9881, 9882, 9883, 9884, 9885, 9886, 9887, 9888, 9889, 9890, 9891, 9892, 9893, 9894, 9895, 9896, 9897, 9898, 9899, 9900, 9901, 9902, 9903, 9904, 9905, 9906, 9907, 9908, 9909, 9910, 9911, 9912, 9913, 9914, 9915, 9916, 9917, 9918, 9919, 9920, 9921, 9922, 9923, 9924, 9925, 9926, 9927, 9928, 9929, 9930, 9931, 9932, 9933, 9934, 9935, 9936, 9937, 9938, 9939, 9940, 9941, 9942, 9943, 9944, 9945, 9946, 9947, 9948, 9949, 9950, 9951, 9952, 9953, 9954, 9955, 9956, 9957, 9958, 9959, 9960, 9961, 9962, 9963, 9964, 9965, 9966, 9967, 9968, 9969, 9970, 9971, 9972, 9973, 9974, 9975, 9976, 9977, 9978, 9979, 9980, 9981, 9982, 9983, 9984, 9985, 9986, 9987, 9988, 9989, 9990, 9991, 9992, 9993, 9994, 9995, 9996, 9997, 9998, 9999, 10000).

Note – You need license for in-depth scanning

Another query “apache 2.4.49”

The screenshot shows the Shodan search interface. The search bar contains the query 'apache 2.4.49'. The results show 1,394 total results. The top countries are listed: United States (301), India (172), and Germany (172). The product spotlight mentions a new API for Fast IP Lookups. The main result is for IP 172.104.208.100, which is an Apache/2.4.49 (Debian) server in Cedar Knolls, United States. The result details include the IP, location, and a list of open ports (80, 443, 8080, 8443, 9000, 9443, 9500, 9501, 9502, 9503, 9504, 9505, 9506, 9507, 9508, 9509, 9510, 9511, 9512, 9513, 9514, 9515, 9516, 9517, 9518, 9519, 9520, 9521, 9522, 9523, 9524, 9525, 9526, 9527, 9528, 9529, 9530, 9531, 9532, 9533, 9534, 9535, 9536, 9537, 9538, 9539, 9540, 9541, 9542, 9543, 9544, 9545, 9546, 9547, 9548, 9549, 9550, 9551, 9552, 9553, 9554, 9555, 9556, 9557, 9558, 9559, 9560, 9561, 9562, 9563, 9564, 9565, 9566, 9567, 9568, 9569, 9570, 9571, 9572, 9573, 9574, 9575, 9576, 9577, 9578, 9579, 9580, 9581, 9582, 9583, 9584, 9585, 9586, 9587, 9588, 9589, 9590, 9591, 9592, 9593, 9594, 9595, 9596, 9597, 9598, 9599, 9600, 9601, 9602, 9603, 9604, 9605, 9606, 9607, 9608, 9609, 9610, 9611, 9612, 9613, 9614, 9615, 9616, 9617, 9618, 9619, 9620, 9621, 9622, 9623, 9624, 9625, 9626, 9627, 9628, 9629, 9630, 9631, 9632, 9633, 9634, 9635, 9636, 9637, 9638, 9639, 9640, 9641, 9642, 9643, 9644, 9645, 9646, 9647, 9648, 9649, 9650, 9651, 9652, 9653, 9654, 9655, 9656, 9657, 9658, 9659, 9660, 9661, 9662, 9663, 9664, 9665, 9666, 9667, 9668, 9669, 9670, 9671, 9672, 9673, 9674, 9675, 9676, 9677, 9678, 9679, 9680, 9681, 9682, 9683, 9684, 9685, 9686, 9687, 9688, 9689, 9690, 9691, 9692, 9693, 9694, 9695, 9696, 9697, 9698, 9699, 9700, 9701, 9702, 9703, 9704, 9705, 9706, 9707, 9708, 9709, 9710, 9711, 9712, 9713, 9714, 9715, 9716, 9717, 9718, 9719, 9720, 9721, 9722, 9723, 9724, 9725, 9726, 9727, 9728, 9729, 9730, 9731, 9732, 9733, 9734, 9735, 9736, 9737, 9738, 9739, 9740, 9741, 9742, 9743, 9744, 9745, 9746, 9747, 9748, 9749, 9750, 9751, 9752, 9753, 9754, 9755, 9756, 9757, 9758, 9759, 9760, 9761, 9762, 9763, 9764, 9765, 9766, 9767, 9768, 9769, 9770, 9771, 9772, 9773, 9774, 9775, 9776, 9777, 9778, 9779, 9780, 9781, 9782, 9783, 9784, 9785, 9786, 9787, 9788, 9789, 9790, 9791, 9792, 9793, 9794, 9795, 9796, 9797, 9798, 9799, 9800, 9801, 9802, 9803, 9804, 9805, 9806, 9807, 9808, 9809, 9810, 9811, 9812, 9813, 9814, 9815, 9816, 9817, 9818, 9819, 9820, 9821, 9822, 9823, 9824, 9825, 9826, 9827, 9828, 9829, 9830, 9831, 9832, 9833, 9834, 9835, 9836, 9837, 9838, 9839, 9840, 9841, 9842, 9843, 9844, 9845, 9846, 9847, 9848, 9849, 9850, 9851, 9852, 9853, 9854, 9855, 9856, 9857, 9858, 9859, 9860, 9861, 9862, 9863, 9864, 9865, 9866, 9867, 9868, 9869, 9870, 9871, 9872, 9873, 9874, 9875, 9876, 9877, 9878, 9879, 9880, 9881, 9882, 9883, 9884, 9885, 9886, 9887, 9888, 9889, 9890, 9891, 9892, 9893, 9894, 9895, 9896, 9897, 9898, 9899, 9900, 9901, 9902, 9903, 9904, 9905, 9906, 9907, 9908, 9909, 9910, 9911, 9912, 9913, 9914, 9915, 9916, 9917, 9918, 9919, 9920, 9921, 9922, 9923, 9924, 9925, 9926, 9927, 9928, 9929, 9930, 9931, 9932, 9933, 9934, 9935, 9936, 9937, 9938, 9939, 9940, 9941, 9942, 9943, 9944, 9945, 9946, 9947, 9948, 9949, 9950, 9951, 9952, 9953, 9954, 9955, 9956, 9957, 9958, 9959, 9960, 9961, 9962, 9963, 9964, 9965, 9966, 9967, 9968, 9969, 9970, 9971, 9972, 9973, 9974, 9975, 9976, 9977, 9978, 9979, 9980, 9981, 9982, 9983, 9984, 9985, 9986, 9987, 9988, 9989, 9990, 9991, 9992, 9993, 9994, 9995, 9996, 9997, 9998, 9999, 10000).

Tools Summary (Student Safe Tools)

Purpose	Tool
Domain info	WHOIS
DNS	dig, nslookup
Infrastructure	Shodan
Tech stack	Wappalyzer
Documents	Google Dorks
Metadata	ExifTool
Social intel	LinkedIn
Code search	GitHub